



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (AI&ML) (CBCS) Degree Examinations
 December 2023

AM1104-1 – INTRODUCTION TO MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

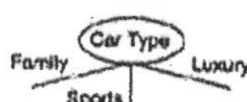
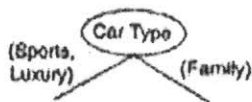
1. **What is the difference between supervised and unsupervised learning?**
 - A) Supervised learning requires labeled data while unsupervised learning does not.
 - B) Unsupervised learning requires labeled data while supervised learning does not.
 - C) Supervised learning does not require data while unsupervised learning does.
 - D) There is no difference between supervised and unsupervised learning.
2. **What is the primary goal of supervised machine learning?**
 - A) Classification
 - B) Clustering
 - C) Dimensionality reduction
 - D) Reinforcement learning
3. **Which type of machine learning algorithm is suitable for predicting a continuous numeric output?**
 - A) Classification
 - B) Clustering
 - C) Regression
 - D) Reinforcement learning
4. **Which of the following techniques is used to handle missing data in a dataset?**
 - A) Removing rows with missing data
 - B) Replacing missing values with zeros
 - C) Imputing missing values with statistical measures
 - D) Ignoring missing data during model training
5. **ML is a field of AI consisting of learning algorithms that?**
 - A) Improve their performance
 - B) At executing some task
 - C) Over time with experience
 - D) All of the above
6. **What is the primary purpose of feature scaling in machine learning?**
 - A) To increase the dimensionality of the dataset
 - B) To make all features have the same scale or range
 - C) To introduce noise into the dataset
 - D) To remove outliers from the dataset
7. **Which of the following machine learning algorithms is considered a non-parametric method?**
 - A) Linear Regression
 - B) Decision Trees
 - C) Support Vector Machines
 - D) K-Nearest Neighbors (KNN)
8. **What is the purpose of feature scaling in machine learning?**
 - A) To convert categorical features into numerical features
 - B) To reduce the dimensionality of the feature space
 - C) To standardize the range of numerical features
 - D) To introduce new features into the model
9. **What is association rule mining used for?**
 - A) Classification
 - B) Regression
 - C) Finding patterns in data
 - D) Natural language processing
10. **Which algorithm is commonly used for association rule mining?**
 - A) K-Means
 - B) Decision Trees
 - C) Apriori
 - D) Linear Regression
11. **In association rule mining, what is support?**
 - A) The confidence of a rule
 - B) The number of transactions containing a rule
 - C) The lift of a rule
 - D) The number of items in a transaction
12. **What is the main goal of Apriori algorithm in association rule mining?**
 - A) Maximize support
 - B) Maximize confidence
 - C) Find frequent itemsets
 - D) Optimize computational efficiency

13. Which of the following is an example of an unsupervised learning algorithm?
 A) Linear Regression B) Decision Trees
 C) K-Means Clustering D) Support Vector Machines
14. Which clustering algorithm does not require specifying the number of clusters beforehand?
 A) K-Means B) Hierarchical Clustering
 C) PCA D) DBSCAN
15. Which clustering algorithm forms clusters by recursively dividing data into smaller sub-clusters?
 A) K-Means B) Hierarchical Clustering
 C) DBSCAN D) Mean-Shift Clustering
16. What is the primary drawback of hierarchical clustering?
 A) Sensitivity to outliers B) Difficulty in determining the number of clusters
 C) Requires a distance metric D) Computationally expensive for large datasets
17. What does ROC stand for in ROC Curve?
 A) Receiver Operating Characteristic B) Random Output Classifier
 C) Regression of Optimal Classification D) Recursive Output Calculation
18. PCA is used for:
 A) Image classification. B) Dimensionality reduction
 C) Natural language processing D) Time series forecasting
19. factor Analysis assumes that the observed variables are influenced by:
 A) A single unobservable factor B) Multiple unobservable factors
 C) No unobservable factors D) The mean of the dataset
20. How is the True Positive Rate (TPR) calculated in an ROC Curve?
 A) $TPR = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$ B) $TPR = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$
 C) $TPR = \frac{\text{True Positives}}{\text{Total Positives}}$ D) $TPR = \frac{\text{True Positives}}{\text{True Positives} + \text{True Negatives}}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Discuss the types of classification techniques in machine learning with an example. | 8 | L*2 | 1 | 1 |
| b) Discuss the different types of Ordered data. | 4 | L2 | 1 | 1 |
| c) Define the following terms: i) Error ii) Accuracy iii) Bias iv) Precision. | 4 | L1 | 1 | 1 |
| 2. a) What are the characteristics of Decision tree algorithm. | 8 | L2 | 2 | 1 |
| b) Compute gini index and entropy for the following nominal attribute CarType. | | | | |



Car Type		
	(Sports, Luxury)	(Family)
C0	9	1
C1	7	3

Car Type		
	(Sports)	(Family, Luxury)
C0	8	2
C1	0	10

Car Type			
	Family	Sports	Luxury
C0	1	8	1
C1	3	0	7

- | | | | | |
|---|---|----|---|---|
| 3. a) With neat diagram explain the General Approach to Solving a Classification Problem. | 8 | L2 | 2 | 1 |
| b) Write an algorithm to implement KNN algorithm. | 4 | L2 | 2 | 1 |
| c) For the confusion matrix, calculate the Accuracy and Error. | | | | |

		Predicted Class	
		Class 1	Class 0
Actual Class	Class 1	45	32
	Class 0	15	23

4	L3	2	2
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Unit – II

4. a) Explain association analysis and apply Apriori algorithm to find frequent 3 itemset for the following data with minimum support of 60%.

Tid	Bread	Milk	Chocolates	Apple	Eggs	Cola
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

- b) Describe linear regression and multi-linear regression.

8 L3 3 2
8 L3 3 2

- 5 a) Discuss how to fix Support vector machine in machine learning.
b) For the following dataset, compute frequent itemsets with minimum support of 40% using Apriori algorithm. Compute confidence of rules. Extract strong rules with minimum confidence=60%.

TID	Items
1	{a, d, e}
2	{a, b, c, e}
3	{a, b, d, e}
4	{a, c, d, e}
5	{b, c, e}
6	{b, d, e}

8 L2 3 1

6. a) Explain DBSCAN algorithm with an example.
b) For the given proximity matrix, construct dendrogram and nested cluster representation using single link (MIN) agglomerative hierarchical clustering (AHC) method.

	1	2	3	4	5
1	0				
2	9	0			
3	3	7	0		
4	6	5	9	0	
5	11	10	2	8	0

8 L3 3 2

8 L2 4 1

8 L3 4 2

Unit – III

7. a) Explain dimensionality reduction using PCA.
b) Illustrate the ensemble techniques of random forest algorithm.
8. a) Describe class imbalance problem.
b) Explain in detail ROC curve and its use in machine learning.

8 L2 5 1

8 L2 5 1

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
